JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY , NOIDA

B.TECH SEMESTER VI

MINOR PROJECT - II (15B19EC691)

Performance Analysis of Lead-Free Single Perovskite $MASnI_3$ and Double Perovskite Cs_2BiI_6

Submitted by :

S.No.	Enrollment No.	Name	Batch
1.	23118037	Ankit Kumar Jha	A16

Under the Supervision of :

Mr. Atul Kumar Srivastava

Department of ECE

JIIT, Sector 62, Noida

Submitted to :-

1. Dr. Gaurav Verma, &

2. Dr. Abhijeet Upadhyay

Department of ECE

JIIT, Sector 62, Noida

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CERTIFICATE

This is to certify that the work titled **“Performance Analysis of Lead-Free Single Perovskite $MASnI_3$ and Double Perovskite Cs_2BiI_6”** submitted by “ Ankit Kumar Jha (23118037) , Kartikey Verma (23118020) , and Sarthak Pratap Singh (23118034) ” in partial fulfilment for the award of degree of B. Tech. of Jaypee Institute of Information Technology, Noida has been carried out under my supervision. This work has not been submitted anywhere other than the institute.

(Signature of Supervisor)

Mr. Atul Kumar Srivastava

Department of ECE

JIIT ,Sector 62 ,Noida

Date :

Place :

(Signature of HOD)

Prof. Jitendra Mohan

Department of ECE

JIIT ,Sector 62 ,Noida

Date :

Place :

DECLARATION BY STUDENT's

We declare that this work was a written submission that represents our ideas in our own words and in where others ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Signature of the Student ……………………………………….
Name of the Student - Ankit Kumar Jha
Enrol. No. - 23118037

Signature of the Student ……………………………………….
Name of the Student - Kartikey Verma
Enrol. No. - 23118020

Signature of the Student ……………………………………….
Name of the Student - Sarthak Pratap Singh
Enrol. No. - 23118034

Date :
Place :

ACKNOWLEDGEMENT

We take this opportunity to express our sincere gratitude to our supervisor Mr. Atul Kumar Srivastava, Department of Electronics and Communication Engineering, for his constant guidance, invaluable suggestions, and continuous encouragement throughout the course of our minor project titled **"Performance Analysis of Lead-Free Single Perovskite $MASnI_3$ and Double Perovskite Cs_2BiI_6".** His valuable insights and continuous support have been instrumental in enhancing our technical knowledge and improving the quality of our work. We also acknowledge the collective efforts of our team members, whose dedication, cooperation, and commitment towards the project ensured its successful completion. Working together as a team not only helped us in dividing the responsibilities effectively but also enriched our learning experience through collaboration and knowledge sharing.

Abstract

In this performance the analysis of lead free perovskite solar has been done. Parameters like Voc, Jsc, FF, and efficiency variation with the absorber layer thickness variation have been performed. Electron Transport Layer (ETL), Hole Transport Layer(HTL), Series Resistance (Rs), Shunt Resistance(Rsh), Temperature and Defect Density(Nt) of the structure has also been varied and analysed. In this analysis lead free single perovskite MaSnI3 and double perovskite Cs2AgBiI6 has been considered. The optimized perovskite solar cell structure for MaSnI3 is TCO/TiO2/MaSnI3/Cu2O and the obtained values for Voc(V) is 1.0999, Jsc(ma/cm2) is 30.9202, FF(%) is 83.24, and Efficiency(%) is 28.31. And optimized perovskite solar cell structure for Cs2AgBiI6 is TCO/TiO2/Cs2AgBiI6/Cu2O and the obtained values for Voc(V) is 1.848, Jsc(ma/cm2) is 24.19, FF(%) is 82.80, and Efficiency(%) is 23.74. The analysis has been performed on the Solar Cell Capacitance Simulator(SCAPS-1D) developed by Marc Burgelman at Ghent University in the early 2000s.

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1.Introduction

1.1 Origin

The origin of the solar cell can be traced back to 1839, when Alexandre Edmond Becquerel first discovered the photovoltaic effect, demonstrating that light can generate an electric current when it falls on certain materials. This discovery laid the fundamental scientific basis for converting solar energy into electrical energy. Later, in 1873, Willoughby Smith observed that selenium exhibits photoconductivity, meaning its electrical conductivity changes when exposed to light. Building on this observation, Charles Fritts developed the first practical solar cell in 1883 by coating selenium with a thin layer of gold. However, the efficiency of this early device was less than 1%, which limited its practical applications.

A major theoretical advancement came in 1905 with the work of Albert Einstein, who explained the photoelectric effect and provided a clear understanding of how light energy can be converted into electrical energy. This explanation played a crucial role in guiding future research and improving photovoltaic materials. The real breakthrough in solar cell technology occurred in 1954 at Bell Labs, where scientists developed the first efficient silicon-based solar cell with an efficiency of about 6%. This marked the beginning of modern solar technology and enabled its use in practical applications such as space satellites, where solar panels became a reliable source of power.

Following this, solar cell technology evolved significantly over time. The first generation of solar cells was based on crystalline silicon, including monocrystalline and polycrystalline types. These cells offered high efficiency, durability, and long operational life, but their production involved complex and expensive manufacturing processes.

Despite their advantages, perovskite solar cells still face challenges related to long-term stability, environmental concerns due to lead content, and sensitivity to moisture and temperature. Researchers are actively working to overcome these issues by developing more stable materials and encapsulation techniques. Additionally, tandem solar cells combining perovskite with silicon are being explored to achieve even higher efficiencies beyond the limits of single-junction cells. Overall, the evolution of solar cell technology from early discoveries to modern perovskite-based systems highlights continuous innovation aimed at making solar energy more efficient, affordable, and sustainable for future energy needs.

1.2 Evolution of Solar Cell

The evolution of solar cells can be classified into three major generations based on the technology shown in the diagram. The first generation consists of wafer-based silicon solar cells, including monocrystalline and polycrystalline silicon solar cells. These are the earliest and most widely used technologies, known for their high efficiency, durability, and long operational life. Monocrystalline cells provide better efficiency due to their uniform crystal structure, while polycrystalline cells are more affordable but slightly less efficient. However, both types involve complex and costly manufacturing processes using high-purity silicon.

The second generation focuses on thin-film solar cell technology, which was developed to reduce cost and material usage. This generation includes amorphous silicon (a-Si) thin-film solar cells, cadmium telluride (CdTe) solar cells, and also perovskite-based solar cells in emerging thin-film form. These solar cells are lightweight, flexible, and easier to manufacture compared to silicon wafers, making them suitable for large-scale and portable applications. However, their efficiency is generally lower than that of first-generation silicon solar cells.

The third generation represents new and emerging solar technologies aimed at achieving high efficiency with low cost and improved flexibility. This includes nanocrystal-based solar cells, polymer (organic) solar cells, dye-sensitized solar cells (DSSC), and concentrated solar cells. Among them, perovskite solar cells have gained special attention due to their rapid improvement in efficiency and simple fabrication methods. Overall, the evolution from first to third generation shows a clear transition from high-cost, rigid, and stable technologies to low-cost, flexible, and highly efficient emerging solar cell systems.

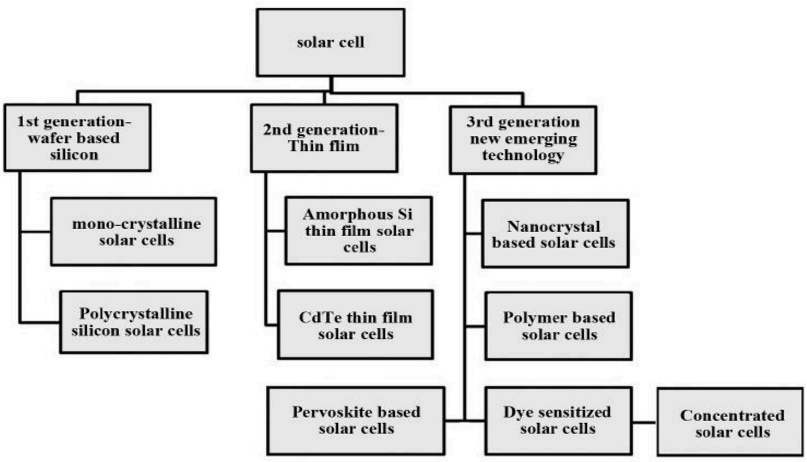

Fig 1:- Evolution of Solar Cell

1.3 Perovskite Solar Cell

A perovskite solar cell (PSC) is a photovoltaic device that uses a perovskite-structured material (such as methylammonium lead halides) as the light-absorbing layer. These materials have excellent properties like high light absorption, long carrier diffusion length, and a tunable bandgap, which enable efficient conversion of sunlight into electricity. When light falls on the perovskite layer, it generates electron–hole pairs; electrons move toward the electron transport layer (ETL) and holes toward the hole transport layer (HTL), producing current.

In addition, perovskite materials allow easy charge separation and transport, which reduces energy loss. Only a thin layer is needed to absorb most of the sunlight, making the device cost-effective. The ETL and HTL are designed to selectively carry electrons and holes, minimizing recombination and improving efficiency. Although perovskite solar cells offer high performance and low-cost fabrication, challenges like stability and moisture sensitivity still need to be addressed for long-term use.

The structure shown in fig. 2 consists of multiple layers, each having a specific role:

1. FTO/ITO Coated Glass (Substrate)

This is the bottom transparent conducting layer. It allows sunlight to pass through to the active layer while also acting as an electrode to collect electrons. Materials like FTO and ITO are used because they are both transparent and electrically conductive.

2. Hole Blocking Layer (HBL)

This thin layer (often compact TiO_2) prevents holes from moving toward the bottom electrode. Its main role is to reduce recombination, ensuring that only electrons are collected at the correct side.

3. Electron Transport Layer (ETL)

The ETL extracts electrons from the perovskite layer and transports them efficiently to the FTO/ITO electrode. It also aligns energy levels properly, which helps in faster electron movement and reduced energy loss. Common materials include TiO_2 and SnO_2.

4. Perovskite Layer (Active Layer)

This is the heart of the solar cell. When sunlight strikes this layer, photons are absorbed and generate electron–hole pairs (charge carriers). Due to the material's properties, these charges are created and separated very efficiently, which is why PSCs show high performance.

5. Hole Transport Layer (HTL)

The HTL collects and transports holes (positive charges) from the perovskite layer to the top electrode. It also blocks electrons, ensuring proper charge separation. Materials like Spiro-OMeTAD or PEDOT:PSS are commonly used.

6. Metallic Counter Electrode

This is the top electrode (gold, silver, or aluminum). It collects holes from the HTL and completes the circuit, allowing current to flow through an external load.

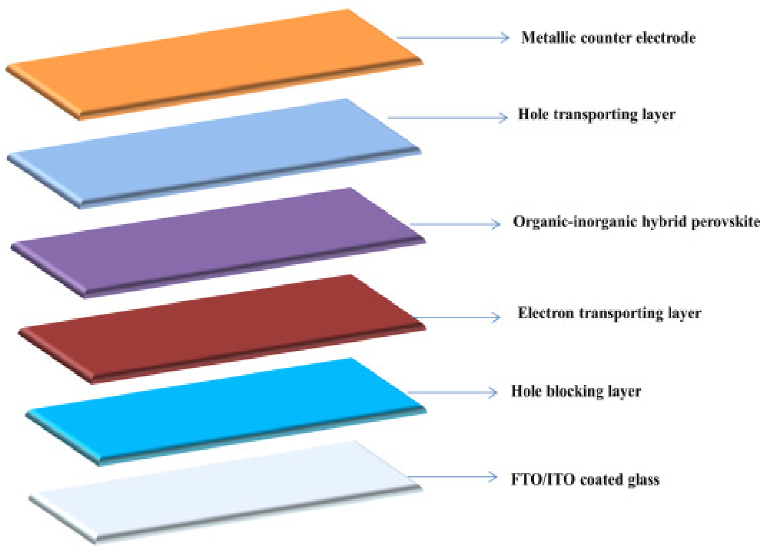

Fig. 2-Structure of Perovskite Solar Cell

1.4 Problem Statement

Perovskite solar cells have shown excellent performance, but the use of toxic lead materials limits their practical applications. To solve this issue, lead-free materials like $MASnI_3$ and Cs_2AgBiI_6 are being studied.

$MASnI_3$ provides good efficiency but suffers from poor stability, which reduces its lifetime. On the other hand, Cs_2AgBiI_6 is stable and safe but gives lower efficiency. This creates a challenge in selecting the best material for solar cell applications.

The main problem is to compare these two materials and understand which one performs better under different conditions. It is also important to study how different parameters like thickness, resistance, and material properties affect the overall performance of the solar cell.

This project aims to analyze and compare $MASnI_3$ and Cs_2AgBiI_6 using simulation methods and find ways to improve their performance.

1.5 Gantt Chart

Phase	Work	Duration
1.	Literature Review	Week 1
2.	Learn SCAPS Tool	Week 2-4
3.	Performance the Simulation	Week 5-8
4.	Result Analysis	Week 9-10
5.	Report and PPT	Week 11-12

1.6 Scope

This study focuses on the performance analysis of lead-free perovskite solar cells using $MASnI_3$ and Cs_2AgBiI_6 materials. The main goal is to compare their efficiency, stability, and overall behavior.

The study includes:

- Comparison of two materials ($MASnI_3$ and Cs_2AgBiI_6)
- Analysis of important parameters like Voc, Jsc, Fill Factor, and Efficiency
- Study the effect on changing the ETL and HTL material
- Study of the effect of resistance and thickness on performance
- Study the effect of temperature and defect density
- Use of simulation tools for analysis

This work is limited to simulation only and does not include practical experiments. The results will help in understanding which material is better and how their performance can be improved in future research.

2. Literature Survey

2.1. Overview

In recent years, perovskite solar cells have become one of the most promising technologies for next-generation solar energy. Many researchers have worked on improving their efficiency and stability. Earlier studies mainly focused on lead-based perovskites, which achieved very high efficiencies. However, due to the toxic nature of lead, researchers started exploring lead-free alternatives.

One of the commonly studied lead-free materials is $MASnI_3$ (methylammonium tin iodide). Several studies have shown that $MASnI_3$ has good electrical properties and a suitable bandgap, which helps in achieving high efficiency. It is considered a direct replacement for lead-based materials because of its similar structure. However, researchers have also reported that $MASnI_3$ suffers from poor stability. The main issue is the oxidation of tin (Sn^{2+} to Sn^{4+}), which leads to rapid degradation of the solar cell performance.

To overcome the stability problem, double perovskite materials like Cs_2AgBiI_6 (cesium silver bismuth iodide) have been introduced. Many studies show that Cs_2AgBiI_6 is more stable and environmentally friendly compared to $MASnI_3$. It does not degrade easily and has better chemical stability. However, its efficiency is lower because it has a larger and indirect bandgap, which reduces light absorption and charge transport.

Some researchers have also worked on improving the performance of these materials by changing different parameters such as layer thickness, electron transport layer (ETL), hole transport layer (HTL), and resistances. Simulation tools like SCAPS-1D are widely used to analyze and optimize these parameters.

Overall, previous studies show a trade-off: $MASnI_3$ provides better efficiency but poor stability, while Cs_2AgBiI_6 offers better stability but lower efficiency. Therefore, more research is needed to improve both properties together.

2.2. Limitations

From the literature review, several research gaps and limitations can be identified:

1. **Trade-off between efficiency and stability**
   $MASnI_3$ shows good efficiency but poor stability, while Cs_2AgBiI_6 is stable but less efficient. No material currently provides both.
2. **Limited parameter optimization**
   Effects of parameters like series resistance (Rs), shunt resistance (Rsh), and thickness are not fully explored.
3. **Lack of proper comparison**
   Many studies focus on only one material instead of comparing both under the same conditions.
4. **Simulation-only approach**
   Most work is based on simulations, with limited real-world validation.
5. **Low efficiency in double perovskites**
   Cs_2AgBiI_6 suffers from low efficiency due to indirect bandgap and weaker charge transport.
6. **Less use of advanced techniques**
   AI/ML-based optimization is rarely used in existing studies.

This project focuses on comparing $MASnI_3$ and Cs_2AgBiI_6 and analyzing how different parameters affect their performance, helping to improve efficiency and stability.

3.Methodology

3.1. Proposed Methodology

In this project, a simulation-based approach is used to analyze and compare the performance of two lead-free perovskite materials: $MASnI_3$ and Cs_2AgBiI_6. The main goal is to study their efficiency, stability, and overall solar cell performance.

The methodology follows a step-by-step process:

1. **Material Selection**
   Two lead-free materials, $MASnI_3$ and Cs_2AgBiI_6, are selected based on their importance in recent research.
2. **Device Structure Design**
   A standard solar cell structure is designed:
   Glass / FTO / ETL / Perovskite Layer / HTL / Metal Contact
3. **Parameter Definition**
   Important parameters such as bandgap, thickness, electron mobility, hole mobility, series resistance (Rs), and shunt resistance (Rsh) are defined for both materials.
4. **Simulation Setup**
   The solar cell is modeled using simulation software, and all material parameters are entered.
5. **Performance Analysis**
   Key output parameters are analyzed:
   - Open Circuit Voltage (Voc)
   - Short Circuit Current Density (Jsc)
   - Fill Factor (FF)
   - Efficiency (η)
6. **Parameter Variation**
   Different values of Rs, Rsh, and thickness are varied to observe their effect on performance.
7. **Comparison of Results**
   The results of $MASnI_3$ and Cs_2AgBiI_6 are compared to identify which material performs better.

Flowchart

Start

↓

Material Selection ($MASnI_3$, Cs_2AgBiI_6)

↓

Device Structure Design

↓

Input Parameters Setup

↓

Simulation Execution

↓

Result Analysis (Voc, Jsc, FF, Efficiency)

↓

Parameter Variation (Rs, Rsh, Thickness)

↓

Comparison of Materials

↓

Conclusion

3.2. Tools and Technique

In this project, simulation tools and analysis techniques are selected carefully to ensure accurate and efficient results.

1. Simulation Tool (SCAPS-1D)

SCAPS-1D is used for modeling and simulation of solar cells.

Why SCAPS-1D is chosen:

- It is specially designed for solar cell analysis
- Easy to use and widely accepted in research
- Allows detailed parameter input

- Provides accurate results for Voc, Jsc, FF, and efficiency
- Supports layer-by-layer device structure

2. Technique Used: Simulation-Based Analysis

Instead of practical fabrication, a simulation-based method is used.

Reasons for choosing simulation:

- Saves time and cost
- No need for physical materials or lab setup
- Easy to test multiple conditions
- Allows quick comparison between materials

3. Parameter Variation Technique

Different parameters like Rs, Rsh, and thickness are varied.

Why this technique is important:

- Helps in understanding how each parameter affects performance
- Useful for optimization of solar cell design
- Gives better insight into material behavior

4. Comparative Analysis Technique

Both materials ($MASnI_3$ and Cs_2AgBiI_6) are analyzed under the same conditions.

Why comparison is needed:

- To find the better material
- To understand trade-offs between efficiency and stability
- To support final conclusions with clear results

4.Implementation & Testing

4.1 Experimental setup

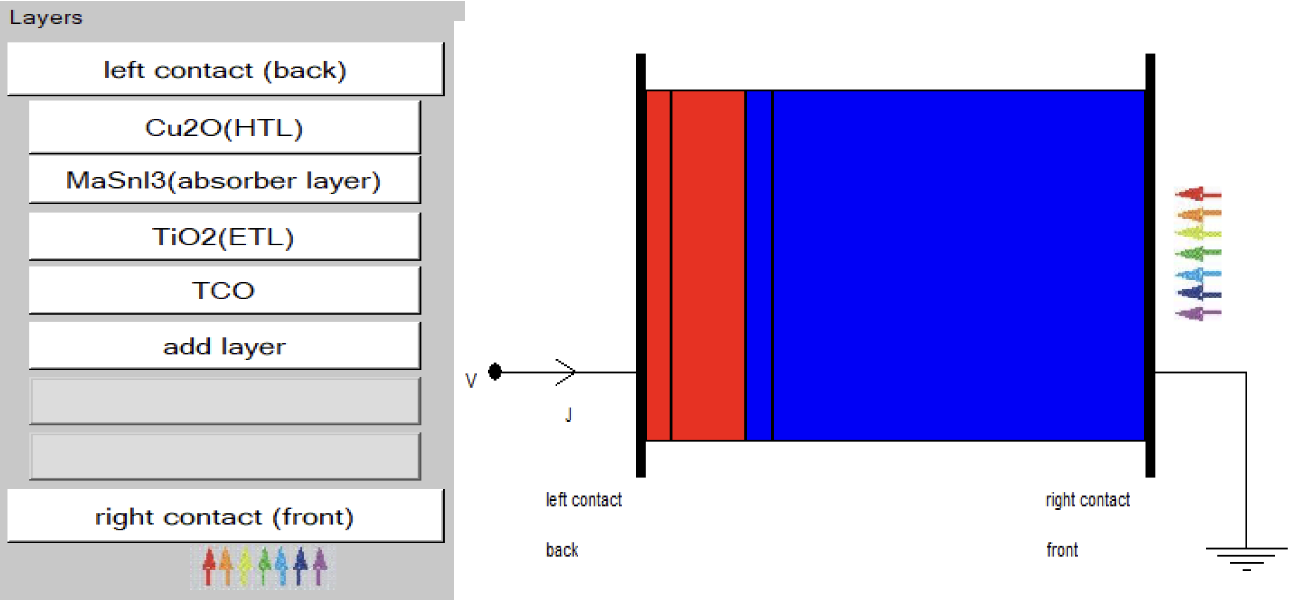

Fig 3:- Schematic Diagram of Perovskite Solar Cell

Device Structure

FTO / ETL / Perovskite / HTL / Metal

Materials Used

- $MASnI_3$ (single perovskite)
- Cs_2AgBiI_6 (double perovskite)

Simulation Conditions

- Temperature = 300 K
- Illumination = 1000 W/m²
- Tool = SCAPS-1D

Table 1 :- Data of different layer of solar cell

Parameters	**Cu2O**	**MaSnI3**	**TiO2**	**TCO**
t (nm)	35	700	35	500
Eg (eV)	2.17	1.4	3.2	3.5
χ (eV)	3.2	4.2	4	4
e_r	7.1	8.2	9	9
$N_c(cm^{-3})$	$2.5*10^{18}$	$1*10^{18}$	$2*10^{18}$	$2.2*10^{18}$
$Nv(cm^{-3})$	$1.8*10^{19}$	$1*10^{18}$	$1.8*10^{19}$	$1.8*10^{19}$
$V_{et}(cm/s)$	$1*10^{7}$	$1*10^{7}$	$1*10^{7}$	$1*10^{7}$
$V_{ht}(cm/s)$	$1*10^{7}$	$1*10^{7}$	$1*10^{7}$	$1*10^{7}$
$\mu_n(cm^2/Vs)$	200	1.6	20	20
$\mu_p\ (cm^2/Vs)$	800	1.6	10	10
$N_A(cm^{-3})$	$1*10^{17}$	$1*10^{16}$	$1*10^{10}$	$1*10^{10}$
$N_D(cm^{-3})$	$1*10^{10}$	$1*10^{10}$	$1*10^{19}$	$1*10^{15}$
$N_T(cm^{-3})$	$1*10^{14}$	$2.5*10^{14}$	$1*10^{14}$	$1*10^{14}$

Table 2 :- Data of different ETL layer of solar cell

Parameter	**TiO2**	**PCBM**	**ZnO:Al**
t (nm)	35	30	50
Eg (eV)	3.2	2	3.25
χ (eV)	4	3.9	4
e_r	9	4	9
$N_c(cm^{-3})$	$2*10^{18}$	$1*10^{20}$	$2*10^{18}$
$Nv(cm^{-3})$	$1.8*10^{19}$	$2*10^{21}$	$1.8*10^{19}$
$V_{et}(cm/s)$	$1*10^{7}$	$1*10^{7}$	$1*10^{7}$
$V_{ht}(cm/s)$	$1*10^{7}$	$1*10^{7}$	$1*10^{7}$
$\mu_n(cm^2/Vs)$	20	$1*10^{-2}$	$3*10^{2}$
$\mu_p\ (cm^2/Vs)$	10	$1*10^{-2}$	$2.5*10^{1}$
$N_A(cm^{-3})$	$1*10^{10}$	$1*10^{10}$	$1*10^{18}$
$N_D(cm^{-3})$	$1*10^{19}$	$1*10^{14}$	$7.25*10^{18}$
$N_T(cm^{-3})$	$1*10^{14}$	$2.5*10^{19}$	0

Table 3 :-Data of different HTL layer of solar cell

Parameter	**Cu2O**	**CuI**	**CuSCN**	**Spiro-OMeTAD**
t (nm)	35	35	100	100
Eg (eV)	2.17	2.98	3.4	3.17
χ (eV)	3.2	2.1	1.9	2.05
e_r	7.1	6.5	6.5	3
$N_c(cm^{-3})$	$2.5*10^{18}$	$2.8*10^{19}$	$1.70*10^{19}$	$2.20*10^{18}$
$Nv(cm^{-3})$	$1.8*10^{19}$	$1*10^{19}$	$2.5*10^{21}$	$1.80*10^{19}$
$V_{et}(cm/s)$	$1*10^{7}$	$1*10^{7}$	$1*10^{7}$	$1*10^{7}$
$V_{ht}(cm/s)$	$1*10^{7}$	$1*10^{7}$	$1*10^{7}$	$1*10^{7}$
$\mu_n(cm^2/Vs)$	200	$1.7*10^{-4}$	$1*10^{-4}$	$2*10^{-4}$
$\mu_p\ (cm^2/Vs)$	800	$1.7*10^{-4}$	$1*10^{-1}$	$2*10^{-4}$
$N_A(cm^{-3})$	$1*10^{17}$	$1*10^{19}$	$1*10^{18}$	$1*10^{19}$
$N_D(cm^{-3})$	$1*10^{10}$	$1*10^{10}$	0	0
$N_T(cm^{-3})$	$1*10^{14}$	$2.5*10^{14}$	$1*10^{14}$	$1*10^{14}$

Table 4 :-Data of different absorber layer of solar cell

Parameter	**MaSnI3**	**Cs2AgBiI6**
t (nm)	700	700
Eg (eV)	1.4	1.6
χ (eV)	4.2	4.05
e_r	8.2	6.5
$N_c(cm^{-3})$	$1*10^{18}$	$1*10^{19}$
$Nv(cm^{-3})$	$1*10^{18}$	$1*10^{19}$
$V_{et}(cm/s)$	$1*10^{7}$	$1*10^{7}$
$V_{ht}(cm/s)$	$1*10^{7}$	$1*10^{7}$
$\mu_n(cm^2/Vs)$	1.6	20
$\mu_p\ (cm^2/Vs)$	1.6	20
$N_A(cm^{-3})$	$1*10^{16}$	$1*10^{14}$
$N_D(cm^{-3})$	$1*10^{10}$	$1*10^{10}$
$N_T(cm^{-3})$	$2.5*10^{14}$	$1*10^{14}$

4.2. Test Cases and Output

Table 5:- When absorber layer thickness of $MASnI_3$ varies

Thickness(nm)	Voc(V)	Jsc(ma/cm2)	FF(%)	ETA(%)
100	1.1762	14.762957	87.28	15.16
200	1.1459	21.629816	82.22	20.38
300	1.1331	25.413275	82.44	23.74
400	1.1246	27.613597	82.96	25.76
500	1.1181	28.884635	83.22	26.88
600	1.1128	29.667865	83.38	27.53
700	1.1086	30.181638	83.42	27.91
800	1.1051	30.525296	83.39	28.13
900	1.1023	30.760716	83.32	28.25
1000	1.0999	30.920212	83.24	28.31

Table 6:- When absorber layer thickness of Cs_2AgBiI_6 aries

Thickness(nm)	Voc(V)	Jsc(ma/cm2)	FF(%)	ETA(%)
100	1.2797	10.919846	88.14	12.32
200	1.2550	16.108573	87.11	17.61
300	1.2399	19.112408	86.13	20.41
400	1.2280	20.933577	85.38	21.95
500	1.2183	22.086938	84.76	22.81
600	1.2098	22.847349	84.24	23.28
700	1.2025	23.367189	83.79	23.55
800	1.1960	23.734330	83.42	23.68
900	1.1901	24.001112	83.10	23.73
1000	1.1848	24.199779	82.80	23.74

Table 7 :- Comparison of photovoltaic performance parameters for different ETL and HTL materials at absorber thickness of 700 nm of $MASnI_3$

Structure	**Voc(V)**	**Jsc(mA/cm2)**	**FF(%)**	**ETA(%)**
TCO/TiO2/MaSnI3/Cu2O	1.10	30.18	83.42	27.91
TCO/TiO2/MaSnI3/CuI	1.10	30.13	69.58	23.25
TCO/TiO2/MaSnI3/CuSCN	1.10	30.14	72.06	24.08
TCO/TiO2/MaSnI3/Spiro-OMeTAD	1.10	30.16	79.91	26.73
TCO/PCBM/MaSnI3/Cu2O	1.10	30.15	83.45	27.89
TCO/PCBM/MaSnI3/CuI	1.10	30.10	69.60	23.23
TCO/PCBM/MaSnI3/CuSCN	1.10	30.11	72.08	24.07
TCO/PCBM/MaSnI3/Spiro-OMeTAD	1.10	30.14	79.93	26.71
TCO/ZnO:Al/MaSnI3/Cu2O	1.10	30.18	83.42	27.91
TCO/ZnO:Al/MaSnI3/CuI	1.10	30.13	69.58	23.25
TCO/ZnO:Al/MaSnI3/CuSCN	1.10	30.13	70.65	23.61
TCO/ZnO:Al/MaSnI3/Spiro-OMeTAD	1.10	30.17	79.90	26.73

Table 8 :- Comparison of photovoltaic performance parameters for different ETL and HTL materials at absorber thickness of 700 nm of Cs_2AgBiI_6

Structure	**Voc(V)**	**Jsc(mA/cm2)**	**FF(%)**	**ETA(%)**
TCO/TiO2/CsAg2BiI6/Cu2O	1.20	23.36	83.79	23.55
TCO/TiO2/CsAg2BiI6/CuI	1.20	23.36	77.66	21.82
TCO/TiO2/CsAg2BiI6/CuSCN	1.20	23.36	80.10	22.51
TCO/TiO2/CsAg2BiI6/Spiro-OMeTAD	1.20	23.36	83.14	23.36

TCO/PCBM/CsAg2BiI6/Cu2O	1.20	23.33	83.79	23.51
TCO/PCBM/CsAg2BiI6/CuI	1.20	23.33	77.65	21.79
TCO/PCBM/CsAg2BiI6/CuSCN	1.20	23.33	80.09	22.48
TCO/PCBM/CsAg2BiI6/Spiro-OMeTAD	1.20	23.33	83.14	23.33
TCO/ZnO:Al/CsAg2BiI6/Cu2O	1.20	23.36	83.79	23.55
TCO/ZnO:Al/CsAg2BiI6/CuI	1.20	23.36	77.65	21.82
TCO/ZnO:Al/CsAg2BiI6/CuSCN	1.20	23.36	80.09	22.51
TCO/ZnO:Al/CsAg2BiI6/Spiro-OMeTAD	1.20	23.36	83.14	23.36

Table 9:- When the series resistance of $MASnI_3$ is changed

Rs(Ω cm2)	**Voc(V)**	**Jsc(ma/cm2)**	**FF(%)**	**ETA(%)**
5	1.1184	28.850999	71.89	23.19
10	1.1184	28.811738	61.05	19.67
15	1.1185	28.764723	51.06	16.43
20	1.1185	28.706451	42.53	13.66
25	1.1185	28.629678	35.09	11.24
30	1.1185	28.489344	30.58	9.74
35	1.1185	27.680782	26.56	8.22
40	1.1185	25.571941	26.86	7.68
45	1.1185	23.294100	26.29	6.85
50	1.1185	21.243960	25.82	6.13

Table 10:- When the series resistance of Cs_2AgBiI_6 is changed

Rs(Ω cm2)	Voc(V)	Jsc(ma/cm2)	FF(%)	ETA(%)
5	1.2184	22.086799	76.36	20.55
10	1.2185	22.086614	68.12	18.33
15	1.2185	22.086350	60.25	16.21
20	1.2185	22.085936	52.86	14.23
25	1.2185	22.085191	46.23	12.44
30	1.2185	22.083534	40.6	10.92
35	1.2185	22.079091	35.64	9.59
40	1.2185	22.067951	32.15	8.64
45	1.2185	21.986415	28.93	7.75
50	1.2185	21.358059	26.73	6.96

Table 11 :- When the shunt resistance of $MASnI_3$ is changed

Rsh(Ω cm2)	Voc(V)	Jsc(ma/cm2)	FF(%)	ETA(%)
500	1.1147	28.884635	77.51	24.96
1000	1.1164	28.884635	80.36	25.91
1500	1.1170	28.884635	81.31	26.24
2000	1.1173	28.884635	81.79	26.40
2500	1.1174	28.884635	82.08	26.49
3000	1.1176	28.884635	82.27	26.56
3500	1.1176	28.884635	82.40	26.60
4000	1.1177	28.884635	82.51	26.64
4500	1.1177	28.884635	82.59	26.66
5000	1.1178	28.884635	82.65	26.68

Table12:- When the shunt resistance of Cs_2AgBiI_6 is changed

Rsh(Ω cm2)	Voc(V)	Jsc(ma/cm2)	FF(%)	ETA(%)
500	1.2125	22.086938	76.68	20.54
1000	1.2155	22.086938	80.71	21.67
1500	1.2164	22.086938	82.06	22.05
2000	1.2169	22.086938	82.73	22.24
2500	1.2172	22.086938	83.14	22.35
3000	1.2174	22.086938	83.41	22.43
3500	1.2175	22.086938	83.6	22.48
4000	1.2176	22.086938	83.74	22.52
4500	1.2177	22.086938	83.86	22.55
5000	1.2177	22.086938	83.95	22.58

Table13:- When the temperature of $MASnI_3$ is changed

Temperature	Voc(V)	Jsc(ma/cm2)	FF(%)	ETA(%)
270	1.1469	30.180116	83.65	28.96
285	1.1279	30.182643	83.61	28.46
300	1.1086	30.184513	83.44	27.92
315	1.089	30.182703	83.15	27.33
330	1.0692	30.183229	82.79	26.72
345	1.0492	30.18367	82.35	26.08
360	1.029	30.184481	81.85	25.42
375	1.0086	30.185338	81.29	24.75
390	0.9879	30.186472	80.68	24.06
405	0.9671	30.188377	80.03	23.36

Table14:-When the temperature of Cs_2AgBiI_6 is changed

Temperature	**Voc(V)**	**Jsc(ma/cm2)**	**FF(%)**	**ETA(%)**
270	1.2517	23.361188	84.68	24.76
285	1.2272	23.361969	84.25	24.16
300	1.2025	23.362811	83.69	23.51
315	1.1775	23.363703	83.04	22.84
330	1.1522	23.364443	82.32	22.16
345	1.1268	23.364916	81.53	21.47
360	1.1012	23.365735	80.69	20.76
375	1.0752	23.366743	79.81	20.05
390	1.0491	23.367786	78.88	19.34
405	1.0228	23.369307	77.90	18.62

Table15 :- When the defect density of $MASnI_3$ is changed

Defect density(Nt)	**Voc(V)**	**Jsc(ma/cm2)**	**FF(%)**	**ETA(%)**
1.00E+10	1.6069	30.505994	86.08	42.19
1.00E+11	1.4877	30.505867	85.28	38.71
1.00E+12	1.3688	30.504597	84.48	35.28
1.00E+13	1.2509	30.491941	84.71	32.31
1.00E+14	1.1426	30.369594	84.67	29.38
1.00E+15	1.0647	29.474970	79.63	24.99
1.00E+16	0.9970	27.692075	71.85	19.84

Table16 :- When the defect density of Cs_2AgBiI_6 is changed

Defect density(Nt)	**Voc(V)**	**Jsc(ma/cm2)**	**FF(%)**	**ETA(%)**
1.00E+10	1.6737	23.368225	86.48	33.82
1.00E+11	1.5546	23.368224	85.75	31.15
1.00E+12	1.4358	23.368214	85.00	28.52
1.00E+13	1.3178	23.368121	84.32	25.97
1.00E+14	1.2025	23.367189	83.79	23.55
1.00E+15	1.0949	23.357868	82.56	21.11
1.00E+16	1.0080	23.264954	73.62	17.26

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Chapter 5: Results & Discussion

5.1 Result Analysis

In this study, the performance of lead-free perovskite solar cells based on $MASnI_3$ and Cs_2AgBiI_6 is analyzed using simulation by varying important device parameters such as absorber thickness, electron transport layer (ETL), hole transport layer (HTL), series resistance (Rs), shunt resistance (Rsh), defect density (Nt), and temperature. The key performance parameters considered are open-circuit voltage (Voc), short-circuit current density (Jsc), fill factor (FF), and efficiency (η).

Effect of Absorber Thickness

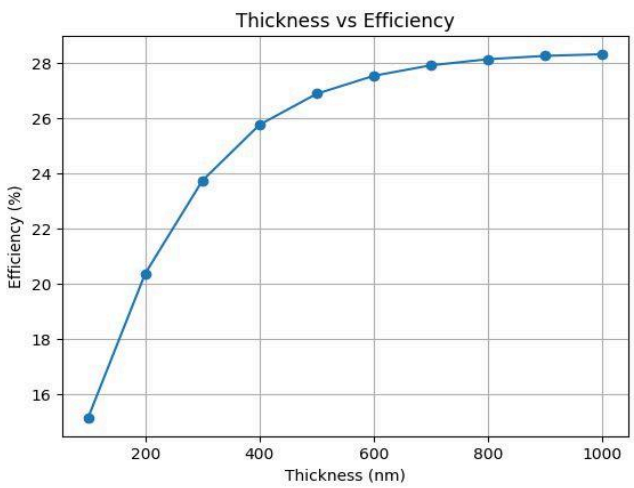


Fig4(a):- ETA Vs Thickness ($MASnI_3$)

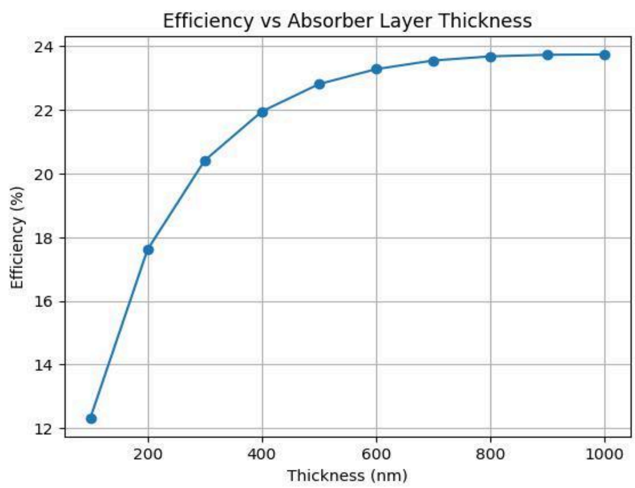


Fig4(b):- ETA Vs Thickness (Cs_2AgBiI_6**)**

The variation of absorber layer thickness has a significant influence on the performance of both $MASnI_3$ and Cs_2AgBiI_6-based perovskite solar cells. As the thickness increases from 100 nm to 1000 nm, both materials show improvement in current density and efficiency due to enhanced light absorption.

For $MASnI_3$, the short-circuit current density (Jsc) increases sharply with thickness, indicating strong absorption capability and efficient charge carrier generation. The efficiency also improves significantly with thickness, showing that the material effectively utilizes incident photons. However, the open-circuit voltage (Voc) decreases gradually with increasing thickness due to increased recombination losses as charge carriers travel longer distances. The fill factor

(FF) initially decreases but stabilizes at higher thickness, indicating relatively balanced charge transport at larger thickness values.

In the case of Cs_2AgBiI_6, a similar trend is observed, where Jsc and efficiency increase with thickness. However, the rate of increase is slower compared to $MASnI_3$, which indicates weaker absorption and lower charge carrier generation. The Voc remains higher than $MASnI_3$ at all thickness values due to the larger bandgap of Cs_2AgBiI_6, but it also shows a gradual decrease with thickness. The fill factor shows a continuous decline, indicating increasing recombination and internal losses.

A detailed comparison shows that $MASnI_3$ consistently provides higher Jsc and efficiency at all thickness levels, while Cs_2AgBiI_6 provides higher Voc but lower overall performance. The improvement in performance with thickness is more significant in $MASnI_3$ due to its direct bandgap and better charge transport properties.

It is also observed that at lower thickness, both materials suffer from poor performance due to insufficient light absorption. As thickness increases, performance improves rapidly initially, but the rate of improvement reduces at higher thickness, indicating a saturation effect. This suggests the presence of an optimum thickness range where the balance between absorption and recombination is achieved.

Additionally, $MASnI_3$ shows a more noticeable drop in Voc with thickness, indicating higher sensitivity to recombination, while Cs_2AgBiI_6 shows comparatively smoother variation, reflecting better stability. The fill factor in $MASnI_3$ remains relatively stable at higher thickness, whereas Cs_2AgBiI_6 shows a gradual decline, suggesting increasing internal losses.

Overall, although both materials benefit from increased thickness, $MASnI_3$ outperforms Cs_2AgBiI_6 in terms of current density and efficiency. The higher Voc of Cs_2AgBiI_6 is not sufficient to compensate for its lower current generation. The best performance for both materials is observed at higher thickness (around 1000 nm), where the gain in absorption dominates the recombination losses.

Effect of ETL and HTL

The performance of $MASnI_3$ and Cs_2AgBiI_6-based perovskite solar cells is analyzed by varying different Electron Transport Layer (ETL) materials (TiO_2, PCBM, ZnO) and Hole Transport Layer (HTL) materials (Cu_2O, CuI, CuSCN, Spiro-OMeTAD) at a fixed absorber thickness of 700 nm.

- **Effect of ETL Materials**

From the results, it is observed that changing the ETL material has negligible impact on the overall device performance for both $MASnI_3$ and Cs_2AgBiI_6.

For **$MASnI_3$**, the values of Voc (~1.1 V) and Jsc (~30.1-30.18 mA/cm²) remain almost constant for all ETL materials. Similarly, the fill factor and efficiency show very minor variations when TiO_2, PCBM, and ZnO are used with the same HTL.

For **Cs_2AgBiI_6**, a similar trend is observed. The Voc (~1.2 V) and Jsc (~23.33-23.36 mA/cm²) remain nearly unchanged for different ETLs. The FF and efficiency also show only slight variation.

This indicates that all selected ETL materials provide good electron extraction and proper energy level alignment, resulting in minimal recombination losses at the ETL interface. Therefore, ETL choice does not significantly influence performance when suitable materials are used.

- **Effect of HTL Materials**

In contrast, the HTL material shows a strong influence on device performance.

For $MASnI_3$:

- With **Cu_2O**, the FF is highest (~83.4%) and efficiency reaches ~27.9%, indicating excellent hole transport and minimal recombination.
- With **CuI**, the FF drops significantly (~69.6%), and efficiency reduces to ~23.2%, showing poor charge extraction and higher recombination.
- With **CuSCN**, the FF improves slightly (~70-72%), and efficiency increases to ~24%, but still remains lower than Cu_2O.

- With **Spiro-OMeTAD**, the FF (~79.9%) and efficiency (~26.7%) improve again, indicating better hole mobility and interface quality.

For Cs_2AgBiI_6:

- **Cu_2O** again provides the best performance, with FF (~83.79%) and efficiency (~23.55%).
- **CuI** shows a noticeable drop in FF (~77.65%) and efficiency (~21.8%).
- **CuSCN** gives moderate performance (~80% FF and ~22.5% efficiency).
- **Spiro-OMeTAD** improves FF (~83.14%) and efficiency (~23.3%), similar to Cu_2O but slightly lower.

Effect of Series Resistance (Rs)

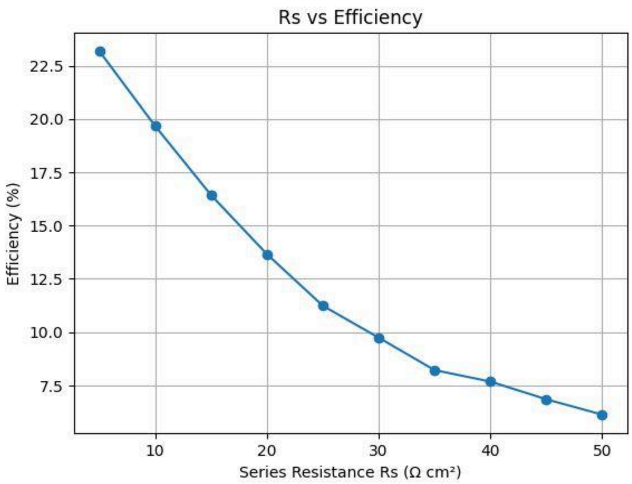


Fig5(a):- ETA Vs Series ($MASnI_3$)

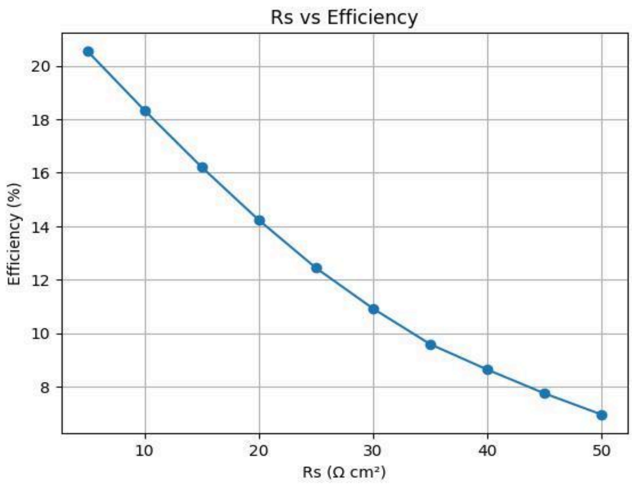


Fig5(b):- ETA Vs Series (Cs_2AgBiI_6)

The effect of series resistance (Rs) on the performance of $MASnI_3$ and Cs_2AgBiI_6-based perovskite solar cells is analyzed over a range from 5 $\Omega \cdot cm^2$ to 50 $\Omega \cdot cm^2$. Series resistance represents internal resistive losses within the device, including contact resistance and bulk resistance, and has a strong influence on fill factor (FF) and efficiency (η).

For **$MASnI_3$**, the open-circuit voltage (Voc) remains almost constant at approximately 1.1185 V across the entire range of Rs, indicating that Voc is not significantly affected by series resistance. However, the short-circuit current density (Jsc) shows a gradual decrease from 28.85 mA/cm² at 5 $\Omega \cdot cm^2$ to about 21.24 mA/cm² at 50 $\Omega \cdot cm^2$ due to increased resistive losses limiting current flow.

The most significant impact is observed in the fill factor (FF), which decreases sharply from 71.89% to about 25.82% as Rs increases. This drastic reduction indicates that higher series resistance severely distorts the J-V characteristics of the solar cell. As a result, the efficiency drops significantly from 23.19% to around 6.13%, showing a major degradation in performance.

For **Cs_2AgBiI_6**, a similar trend is observed. The Voc remains nearly constant at around 1.2185 V, confirming that it is not sensitive to Rs variation. The Jsc remains almost constant initially and then shows a slight decrease at higher Rs values, indicating limited influence on carrier generation.

However, the fill factor decreases significantly from 76.36% at 5 Ω·cm² to about 26.73% at 50 Ω·cm². This reduction is due to increased voltage drop within the device caused by internal resistance. Consequently, the efficiency decreases from 20.55% to around 6.96%.

Effect of Shunt Resistance (Rsh)

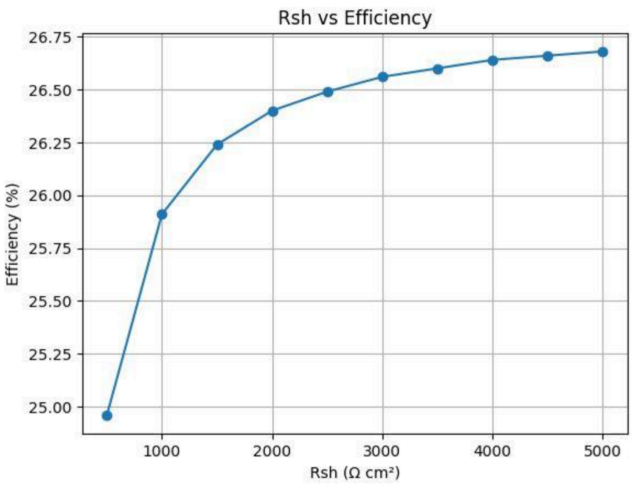


Fig6(a):- ETA Vs Shunt ($MASnI_3$)

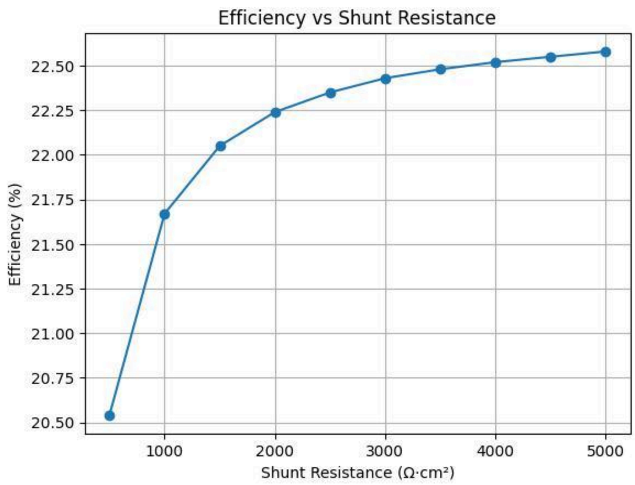


Fig6(b):- ETA Vs Shunt (Cs_2AgBiI_6)

The influence of shunt resistance (Rsh) on the performance of $MASnI_3$ and Cs_2AgBiI_6-based perovskite solar cells is analyzed over the range of 500 Ω·cm² to 5000 Ω·cm². Shunt resistance is an important parasitic parameter that represents leakage current paths within the device, and it plays a major role in determining the fill factor and overall efficiency.

For **$MASnI_3$**, it is observed that the open-circuit voltage (Voc) shows a very small increase from 1.1147 V to 1.1178 V as Rsh increases. This indicates that Voc is almost independent of shunt resistance. Similarly, the short-circuit current density (Jsc) remains constant at approximately 28.88 mA/cm² for all values of Rsh, confirming that charge generation is unaffected by leakage resistance.

However, a significant improvement is observed in the fill factor (FF), which increases from 77.51% at low Rsh (500 Ω·cm²) to 82.65% at high Rsh (5000 Ω·cm²). This increase is due to the reduction in leakage current paths, which allows more efficient extraction of charge carriers without internal loss. As a result, the efficiency improves from 24.96% to 26.68%, showing a clear enhancement in device performance with increasing Rsh.

For **Cs_2AgBiI_6**, a similar trend is observed. The Voc increases slightly from 1.2125 V to 1.2177 V, while Jsc remains constant at approximately 22.09 mA/cm², indicating that photogeneration is not influenced by shunt resistance.

The fill factor shows a strong improvement from 76.68% at 500 Ω·cm² to 83.95% at 5000 Ω·cm². This indicates that Cs_2AgBiI_6 is also highly sensitive to leakage losses at low shunt resistance. Consequently, efficiency increases from 20.54% to 22.58%.

Effect of Defect Density (Nt)

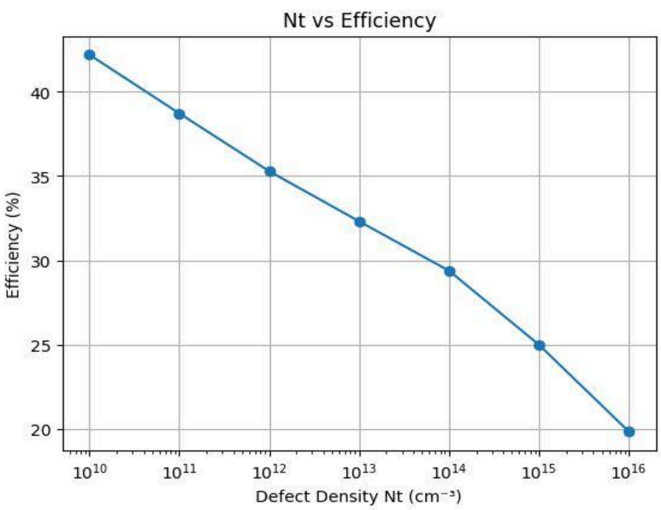


Fig7(a):- ETA Vs Defect ($MASnI_3$)

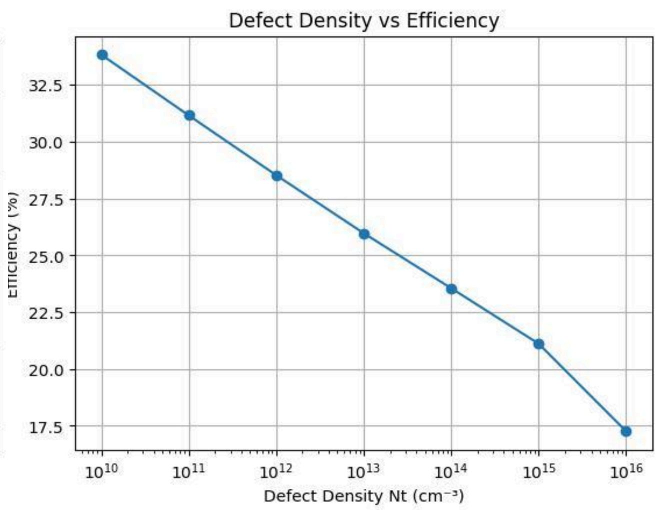


Fig7(b):- ETA Vs Defect(Cs_2AgBiI_6)

The impact of defect density (Nt) on the performance of $MASnI_3$ and Cs_2AgBiI_6 solar cells is analyzed over a range from 10^{10} cm^{-3} to 10^{16} cm^{-3}. It is clearly observed that increasing defect density significantly degrades device performance due to enhanced recombination.

For **$MASnI_3$**, at a very low defect density of 10^{10} cm^{-3}, the device shows excellent performance with a high Voc of 1.6069 V, Jsc of about 30.50 mA/cm², FF of 86.08%, and efficiency of 42.19%. As the defect density increases to 10^{12}–10^{13} cm^{-3}, the efficiency gradually decreases, mainly due to a reduction in Voc, which drops as recombination increases.

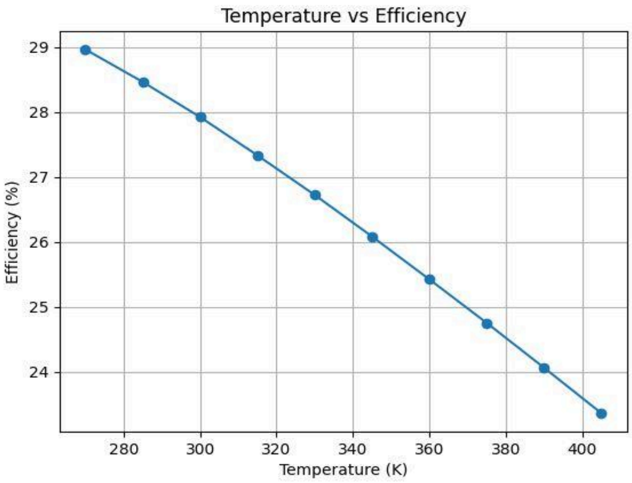
At higher defect densities (10^{14} to 10^{16} cm^{-3}), the performance degrades rapidly. The Voc decreases significantly to 0.997 V at 10^{16} cm^{-3}. The Jsc, which remains nearly constant at lower Nt, starts decreasing noticeably at high defect density, reaching around 27.69 mA/cm². The fill factor also drops sharply from about 86% to nearly 71.85%, indicating severe internal losses. As a result, the efficiency reduces drastically to around 19.84%.

For **Cs_2AgBiI_6**, a similar trend is observed. At low defect density (10^{10} cm^{-3}), the device shows a high Voc of 1.6737 V, Jsc of about 23.36 mA/cm², FF of 86.48%, and efficiency of 33.82%. As Nt increases, the Voc decreases steadily, reaching 1.008 V at 10^{16} cm^{-3}.

The Jsc remains almost constant (~23.36 mA/cm²) up to moderate defect density, indicating that carrier generation is not significantly affected initially. However, at very high Nt (10^{16} cm^{-3}), it slightly decreases due to increased recombination.

The fill factor decreases gradually and then shows a sharp drop at high defect density, reaching around 73.62%. Consequently, the efficiency decreases continuously from 33.82% at 10^{10} cm^{-3} to about 17.26% at 10^{16} cm^{-3}.

Effect of Temperature



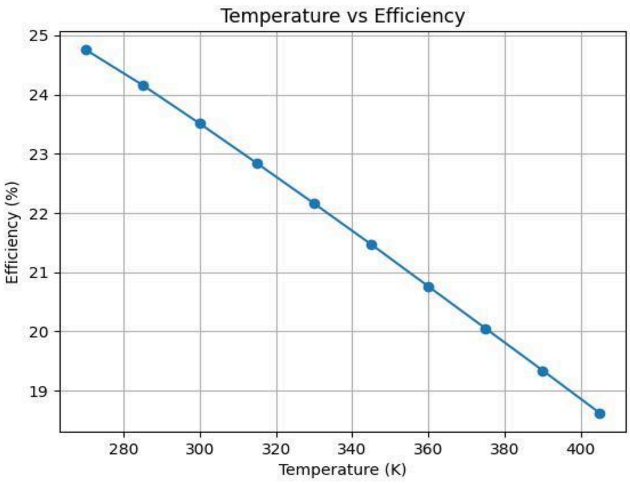


Fig8(a):- ETA Vs Temp. ($MASnI_3$)

Fig8(b):- ETA Vs Temp. (Cs_2AgBiI_6)

The effect of temperature on the performance of $MASnI_3$ and Cs_2AgBiI_6-based perovskite solar cells is analyzed over a range from 270 K to 405 K. It is observed that temperature has a significant impact on all key performance parameters, especially open-circuit voltage (Voc), fill factor (FF), and efficiency (η).

For **$MASnI_3$**, as the temperature increases, the open-circuit voltage (Voc) shows a continuous and noticeable decrease. This reduction is primarily due to the increase in intrinsic carrier concentration and enhanced recombination at higher temperatures. As a result, the built-in potential of the device decreases, leading to lower Voc.

The short-circuit current density (Jsc), however, remains almost constant across the temperature range, with only a very slight increase. This indicates that photon absorption and carrier generation are not significantly affected by temperature.

The fill factor (FF) gradually decreases with increasing temperature due to increased resistive losses and recombination effects. Consequently, the overall efficiency (η) decreases steadily as temperature rises, showing a clear degradation in device performance at higher temperatures.

For **Cs_2AgBiI_6**, a similar trend is observed. The Voc decreases consistently with increasing temperature, but its value remains higher than that of $MASnI_3$ across all temperature ranges due to its larger bandgap.

The Jsc again shows negligible variation, indicating that carrier generation is relatively stable with temperature. However, the fill factor decreases more noticeably compared to $MASnI_3$, indicating higher sensitivity to internal losses.

The efficiency of Cs_2AgBiI_6 also decreases steadily with temperature. The rate of efficiency degradation is slightly higher compared to $MASnI_3$, suggesting that although Cs_2AgBiI_6 is chemically stable, its electrical performance is still affected by thermal conditions.

Overall Observation

The overall performance analysis of $MASnI_3$ and Cs_2AgBiI_6-based perovskite solar cells has been carried out by varying key device parameters, including absorber thickness, ETL, HTL, series resistance (Rs), shunt resistance (Rsh), defect density (Nt), and temperature. Each parameter plays a distinct role in influencing the photovoltaic performance.

The **absorber thickness** significantly affects light absorption and carrier generation. As thickness increases, the short-circuit current density (Jsc) and efficiency improve due to enhanced photon absorption. However, excessive thickness leads to increased recombination losses, causing a reduction in open-circuit voltage (Voc) and slight degradation in fill factor (FF). Thus, an optimal thickness is required to balance absorption and recombination.

The variation of **ETL materials** shows minimal impact on device performance, as all selected ETLs provide proper energy band alignment and efficient electron extraction. In contrast, the **HTL materials** strongly influence performance, particularly the fill factor and efficiency. HTLs with better hole mobility and interface quality result in improved charge extraction and reduced recombination losses.

The **series resistance (Rs)** has a strong negative impact on performance. As Rs increases, resistive losses increase, leading to a significant reduction in fill factor and efficiency. The open-circuit voltage remains nearly unaffected, but overall power output decreases sharply due to internal voltage drop.

On the other hand, increasing the **shunt resistance (Rsh)** improves device performance. Higher Rsh reduces leakage current paths, resulting in improved fill factor and efficiency. The effect on Voc and Jsc is minimal, but overall performance improves due to better charge utilization.

The **defect density (Nt)** plays a critical role in recombination behavior. At low defect density, the device exhibits high efficiency due to minimal recombination. As Nt increases, recombination centers increase, leading to a significant reduction in Voc, FF, and efficiency. At very high defect density, even Jsc is affected due to severe carrier trapping.

The **temperature** variation shows that increasing temperature degrades device performance. Higher temperature increases carrier recombination and reduces the built-in potential, leading to a decrease in Voc and efficiency. The Jsc remains relatively stable, but overall efficiency decreases due to the reduction in Voc and FF.

5.2 Performance Evaluation

The performance of $MASnI_3$ and Cs_2AgBiI_6 is evaluated and compared with existing research on lead-free perovskite solar cells.

From the analysis, $MASnI_3$ shows higher efficiency due to its smaller bandgap and better charge transport properties. It allows higher absorption of light and efficient carrier movement, which results in improved Jsc and overall performance.

However, $MASnI_3$ suffers from stability issues due to oxidation of tin, which is also reported in existing studies. This limits its practical application despite its high efficiency.

On the other hand, Cs_2AgBiI_6 shows lower efficiency due to its larger and indirect bandgap, which reduces light absorption and charge transport. However, it provides better chemical stability and is environmentally safe.

Based on the input parameters, it is clear that the performance difference between the two materials is mainly due to their intrinsic material properties rather than external conditions.

Compared to traditional lead-based perovskites, both materials show lower efficiency, but they eliminate toxicity issues, making them more suitable for sustainable and long-term applications.

Final Comparison

- $MASnI_3$ → High efficiency, low stability
- Cs_2AgBiI_6 → Low efficiency, high stability

Overall, this study confirms the commonly observed trade-off between efficiency and stability in lead-free perovskite solar cells and highlights the need for further optimization to achieve both properties in a single material.

6. Conclusion & Future Scope

6.1 Summary

In this project, a detailed performance analysis of lead-free perovskite solar cells using $MASnI_3$ and Cs_2AgBiI_6 materials has been carried out using simulation techniques. The main objective was to compare these two materials based on efficiency, stability, and overall performance.

The solar cell structure was designed using a standard configuration, and simulations were performed under different conditions. Various parameters such as series resistance (Rs), shunt resistance (Rsh), thickness, ETL, HTL, defect density and temperature were varied to study their effect on performance.

The results show that $MASnI_3$ provides higher efficiency and better electrical performance due to its direct bandgap and good charge transport properties. However, it suffers from stability issues. On the other hand, Cs_2AgBiI_6 shows better stability and is environmentally safer, but its efficiency is lower due to its indirect bandgap.

Through this project, a clear comparison between single and double perovskite materials has been achieved. The study helps in understanding the trade-off between efficiency and stability in lead-free perovskite solar cells.

6.2 Impact on Society, Environmental Sustainability, Ethical Issues and Compliance

This project contributes to the development of environmentally friendly solar energy solutions. Traditional perovskite solar cells use lead, which is toxic and harmful to both human health and the environment. By focusing on lead-free materials like $MASnI_3$ and Cs_2AgBiI_6, this work supports safer and more sustainable alternatives.

From a societal perspective, improving solar cell technology can help in increasing the use of renewable energy, reducing dependence on fossil fuels, and lowering carbon emissions.

In terms of environmental sustainability, Cs_2AgBiI_6 is especially important because it is stable and non-toxic, making it suitable for long-term use without environmental risk.

Ethically, the use of non-toxic materials aligns with global standards for safe and responsible technology development. This work supports compliance with environmental regulations and promotes green energy solutions.

From an industry point of view, this research is relevant as companies are actively searching for stable and eco-friendly solar cell materials for commercial use.

6.3 Limitations

Although the project provides useful insights, there are some limitations:

- The study is based only on simulation (SCAPS-1D) and does not include experimental validation
- Real-world factors like temperature variation and material degradation are not fully considered
- Limited material parameters are used, which may not represent actual conditions completely
- Cs_2AgBiI_6 shows low efficiency, and no advanced optimization method is applied to improve it
- The comparison is limited to only two materials
- Life time of the solar cell is very low compared to the Si-based solar cell

6.4 Future Scope

There are several directions for future improvement and research:

- Experimental fabrication of solar cells to validate simulation results
- Use of advanced optimization techniques such as Artificial Intelligence and Machine Learning
- Exploration of new lead-free perovskite materials with better efficiency and stability
- Improvement of Cs_2AgBiI_6 efficiency through doping or material engineering
- Development of tandem solar cells for higher performance
- Study of long-term stability and degradation effects

Future research can focus on achieving both high efficiency and high stability in a single lead-free material, which will make perovskite solar cells more suitable for real-world applications

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